



PIXELRAG: Web Screenshots Beat Text for Retrieval-Augmented Generation

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Abstract

Augmenting large language models (LLMs) with retrieved web text has become a dominant paradigm, yet the web is not natively textual: existing systems depend on complex parsing pipelines that linearize HTML and discard layout, visual structure, and formatting. We introduce **PIXELRAG**, a new retrieval-augmented method that represents websites in their *native visual form* and performs retrieval and reading entirely in pixel space, enabling an end-to-end architecture that eliminates text abstraction. PIXELRAG is, to our knowledge, the first pipeline to operate over a full Wikipedia corpus in this form, scaling to a datastore of 30 million screenshot images with an efficient visual retrieval index. Built on an existing visual embedding model (i.e., Qwen3-VL-Embedding), PIXELRAG further fine-tunes this model on screenshot data with carefully curated contrastive training data. Retrieved screenshots are then fed directly as pixel inputs to a VLM, without intermediate text conversion. PIXELRAG consistently outperforms both no-retrieval and text-based RAG baselines, most surprisingly on widely studied text-centric tasks such as NQ and SimpleQA. It also achieves strong gains on multimodal open-domain QA (e.g., MMSearch), benchmarks over noisy news corpora (e.g., LiveVQA), and agentic benchmarks (e.g., MoNaCo), improving accuracy by up to 18.1% over text-based baselines. Finally, pixel representations enable a new efficiency lever for RAG through image compression, achieving up to $3\times$ token cost reduction at lower resolutions while maintaining accuracy. Our results challenge the necessity of text representations in web retrieval, suggesting that web RAG can operate directly in the web’s native visual form while improving both performance and efficiency. Our code is available at <https://github.com/StarTrail-org/PixelRAG>.

1 Introduction

Retrieval-augmented generation (RAG) has become a dominant paradigm for grounding large language models (LLMs) in external knowledge, powering open-domain question answering, search-augmented agents, and deep-research systems [1–6]. Among retrieval sources, the web is the largest and most diverse knowledge base: an increasing number of LLM applications rely on web retrieval to stay current and factually grounded [7–9].

Text-based RAG assumes access to clean text documents for retrieval and reading, but hinges on a critical, often overlooked step: HTML parsing. Clean text is not free: HTML-to-text extraction

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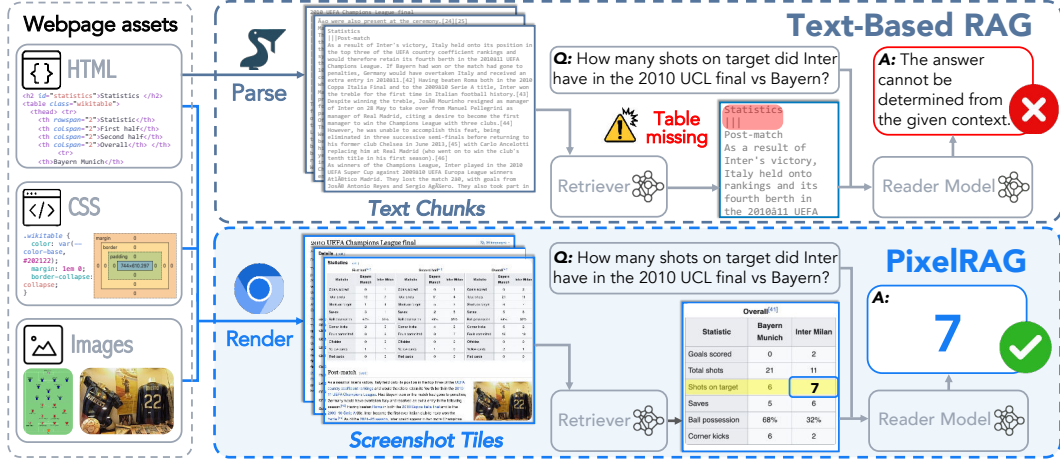


Figure 1: **Overview of PIXELRAG.** Text-based RAG (top) parses HTML into a text index and retrieves text chunks for the reader model. **PIXELRAG** (bottom, ours) renders each webpage, builds a visual index, and retrieves screenshot tiles for the reader — no parser required, fully visual.

is complex, heavily engineered, and error-prone [10–12]. Even state-of-the-art parsers [13–15] are brittle and inherently lossy, discarding visual cues (e.g., emphasis, layout) and structured content such as tables, charts, and images. These losses cascade: the retriever indexes a degraded view of the page, and the reader must reason over linearized, noisy text that may be difficult to comprehend.

In this work, we ask: can RAG operate directly in pixel space—on the web as users see it—bypassing complex and lossy HTML parsing? Concretely, this would index web screenshots, retrieve pages as images, and feed them to a VLM as context. While ambitious, recent advances in VLMs that significantly improved visual understanding, enabling capabilities such as OCR, document, and chart understanding, make this approach feasible [16–21]. As information-seeking increasingly involves structured and visual content, this approach also enables a unified representation without separate text and vision representations. Prior work has explored this paradigm in curated, small-scale domains such as PDFs [22–27], but it remains largely unexplored for the web—the dominant source of open-domain knowledge.

We introduce **PIXELRAG**, to our knowledge, the first end-to-end screenshot-based RAG approach at web scale, operating over the full Wikipedia corpus (7M articles) and news corpora from CNN, AP News, and BBC. While scalable data collection is non-trivial, we build a pipeline that first collects rendering assets (HTML, CSS, images) and reconstructs pages locally to produce screenshots at scale. Retrieval is powered by a Qwen3-VL-Embedding model [28], which we further adapt to screenshot data via efficient contrastive fine-tuning with synthetic QA pairs and dynamic hard-negative mining [29, 3]. Retrieved images are then fed directly as pixel inputs to a VLM reader, bypassing intermediate text representations.

Our experiments show that **PIXELRAG** consistently outperforms no retrieval and text-based RAG baselines across benchmarks, most surprisingly on widely used text-centric Wikipedia QA tasks such as NQ [30] and SimpleQA [31], where most, if not all, questions are answerable from text alone. **PIXELRAG** also delivers strong gains on multimodal QA, benchmarks over noisy news corpora (LiveVQA [32]), and agentic benchmarks [33]. Further analysis suggests that gains on text-centric tasks arise from leveraging 2D structural cues (e.g., tables, infoboxes) and avoiding information loss observed in HTML-to-text parsing. Notably, these gains scale with VLM capability, suggesting further headroom as models continue to improve.

Finally, pixel representations introduce a new, underexplored dimension for improving token efficiency in RAG via image compression, in line with recent work [34, 35]. Training the VLM reader to tolerate lower resolutions further improves the accuracy–efficiency Pareto frontier, enabling up to $3\times$ token cost reduction while maintaining accuracy. To summarize, our contributions are fourfold: